```
M_Homo_sapiens

M_Homo_sapiens                  ..........................................................
A_Catharus_ustulatus            ..........................................................
A_Haemorhous_mexicanus          ..........................................................
A_Phasianus_colchicus           ..........................................................
A_Numida_meleagris              ..........................................................
A_Aythya_fuligula               ..........................................................
A_Cygnus_olor                   ..........................................................
A_Calypte_anna                  ..........................................................
A_Cariama_cristata              ..........................................................
A_Falco_peregrinus              ..........................................................
A_Pygoscelis_adeliae            ..........................................................
A_Opisthocomus_hoazin           ..........................................................
A_Haliaeetus_leucocephalus      ..........................................................
A_Harpia_harpyja                ..........................................................
A_Aquila_chrysaetos_chrysaetos  ..........................................................
A_Rhea_pennata                  ..........................................................
A_Dromaius_novaehollandiae      ..........................................................
M_Ornithorhynchus_anatinus      ..........................................................
M_Sarcophilus_harrisii          ..........................................................
M_Sminthopsis_crassicaudata     ..........................................................
M_Macrotis_lagotis              ..........................................................
M_Phascolarctos_cinereus        ..........................................................
M_Notamacropus_eugenii          ..........................................................
M_Dasypus_novemcinctus          ..........................................................
M_Heterocephalus_glaber         ..........................................................
M_Rattus_norvegicus             ..........................................................
M_Mus_musculus                  ..........................................................
M_Microtus_pennsylvanicus       ..........................................................
M_Phodopus_roborovskii          ..........................................................
M_Cricetulus_griseus            ..........................................................
M_Cavia_porcellus               ..........................................................
M_Oryctolagus_cuniculus         ..........................................................
M_Marmota_monax                 ..........................................................
M_Sciurus_carolinensis          ..........................................................
M_Tenrec_ecaudatus              ..........................................................
M_Loxodonta_africana            ..........................................................
M_Desmodus_rotundus             ..........................................................
M_Camelus_dromedarius           ..........................................................
M_Chrysochloris_asiatica        ..........................................................
M_Bubalus_kerabau               ..........................................................
M_Ovibos_moschatus              ..........................................................
M_Capra_hircus                  ..........................................................
M_Phacochoerus_africanus        ..........................................................
M_Galeopterus_variegatus        ..........................................................
M_Cercocebus_atys               ..........................................................
M_Chlorocebus_sabaeus           ..........................................................
M_Macaca_fascicularis           ..........................................................
M_Pan_troglodytes               ..........................................................
M_Pan_paniscus                  ..........................................MREPLEA.....
M_Gorilla_gorilla               ..........................................MREPLEA.....
M_Orycteropus_afer              ..........................................................
M_Tamandua_tetradactyla         ..........................................................
M_Eschrichtius_robustus         ..........................................................
M_Crocuta_crocuta               ..........................................................
M_Prionailurus_viverrinus       MEGGEGKINQKPLPGARHSLSAFCANASQKPRTCQSPKNFIKMQPDSVAGPEVPHVCKLP
M_Vulpes_lagopus                ......................MCAGG.T..................RS.....WVGP
M_Vulpes_vulpes                 ......................MCAGG.T..................RS.....WVGP
M_Canis_lupus                   ......................MVCDRDSG.................TG.....WPI.
M_Lutra_lutra                   ......................MGSRGG...................AW.....WP..
M_Ursus_maritimus               ..........................................................
M_Halichoerus_grypus            ..........................................................M..
M_Callorhinus_ursinus           ..........................................................
consensus>70                    ..........................................................
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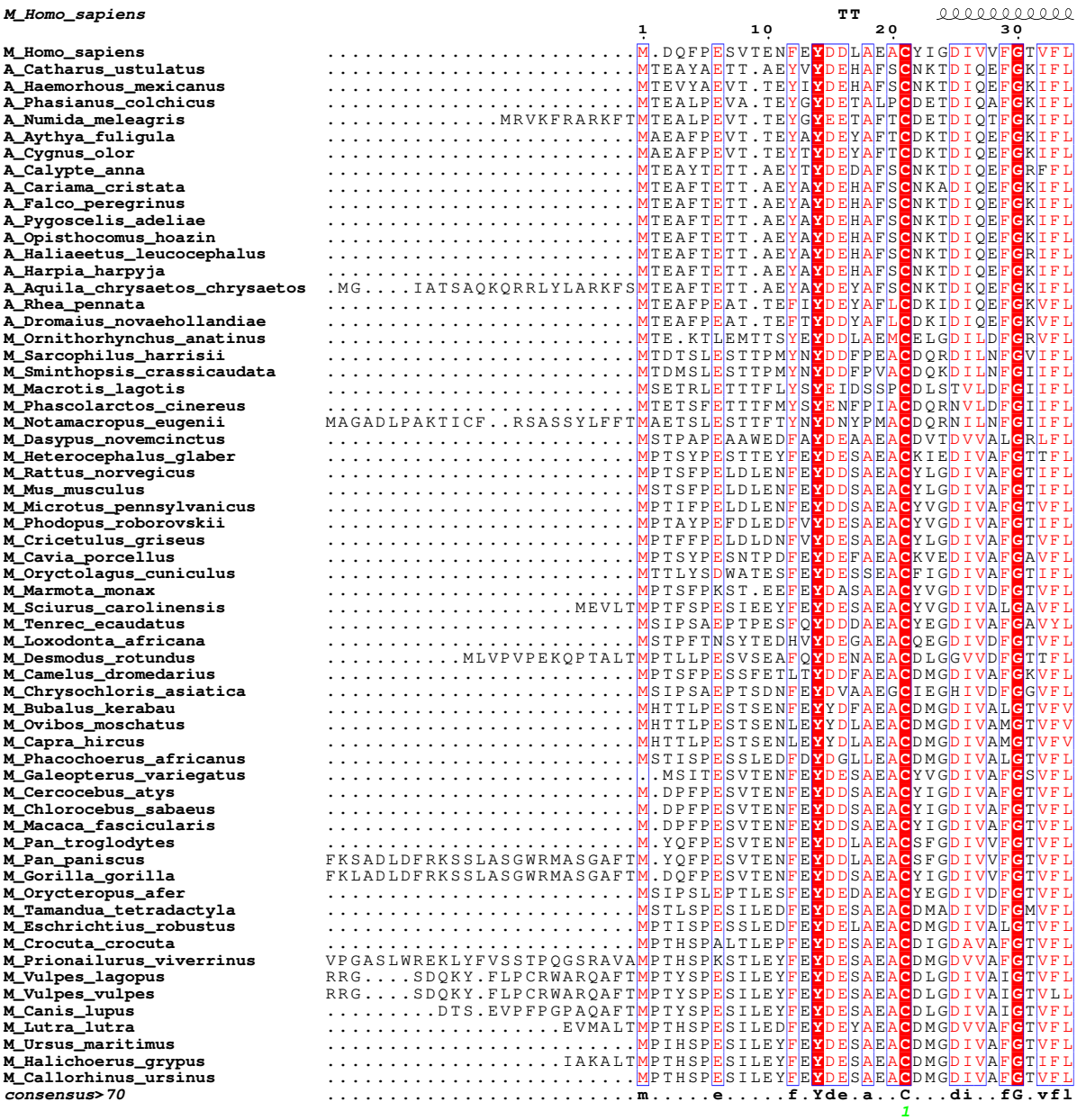
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M_Homo_sapiens
A_Catharus_ustulatus
A_Haemorhous_mexicanus
A_Phasianus_colchicus
A_Numida_meleagris
A_Aythya_fuligula
A_Cygnus_olor
A_Calypte_anna
A_Cariama_cristata
A_Falco_peregrinus
A_Pygoscelis_adeliae
A_Opisthocomus_hoazin
A_Haliaeetus_leucocephalus
A_Harpia_harpyja
A_Aquila_chrysaetos_chrysaetos
A_Rhea_pennata
A_Dromaius_novaehollandiae
M_Ornithorhynchus_anatinus
M_Sarcophilus_harrisii
M_Sminthopsis_crassicaudata
M_Macrotis_lagotis
M_Phascolarctos_cinereus
M_Notamacropus_eugenii
M_Dasypus_novemcinctus
M_Heterocephalus_glaber
M_Rattus_norvegicus
M_Mus_musculus
M_Microtus_pennsylvanicus
M_Phodopus_roborovskii
M_Cricetulus_griseus
M_Cavia_porcellus
M_Oryctolagus_cuniculus
M_Marmota_monax
M_Sciurus_carolinensis
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M_Desmodus_rotundus
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M_Phacochoerus_africanus
M_Galeopterus_variegatus
M_Cercocebus_atys
M_Chlorocebus_sabaeus
M_Macaca_fascicularis
M_Pan_troglodytes
M_Pan_paniscus
M_Gorilla_gorilla
M_Orycteropus_afer
M_Tamandua_tetradactyla
M_Eschrichtius_robustus
M_Crocuta_crocuta
M_Prionailurus_viverrinus
M_Vulpes_lagopus
M_Vulpes_vulpes
M_Canis_lupus
M_Lutra_lutra
M_Ursus_maritimus
M_Halichoerus_grypus
M_Callorhinus_ursinus
consensus>70
1

α2
α3
M_Homo_sapiens
40
50
60
70
80
90
M_Homo_sapiens SIFYSVIFAIGLVGNLLVVFALTNSKKPKSVTDIYLLNLALSDLLFVATLPFWTHYLINE
A_Catharus_ustulatus PIFYIVVFALGLTGNLMVVFAIVK.GNKKSITDIYLLNLAVSDLLFVISLPFWAANTVRG
A_Haemorhous_mexicanus PLFYIAVFALGLTGNLMVVFAIVK.GNKKSITDIYLLNLAISDLLFVISLPFWASNTVRG
A_Phasianus_colchicus PLFYIAVFALGLAGNVMVVLAIVKEGSKKSITDIYLLNLAVSDLLFVISLPFWASNTVRG
A_Numida_meleagris PMFYIAVFALGLAGNVMVVLAIVKEGSKKSITDIYLLNLAVSDLLFVISLPFWASNTVRG
A_Aythya_fuligula PIFYIVVFALGLMGNLMVVFAILKAGSKKSITDIYLLNLAVSDLLFVVSLPFWASNTVRG
A_Cygnus_olor PIFYIVMFALGLTGNLMVVFAILKAGSKKSITDIYLLNLAVSDLLFVVSLPFWASNTVRG
A_Calypte_anna PIFYLAVFALGLTGNIMVVFAIAKEGNKKSITDIYLLNLAVSDLLFVISLPFWASNTVRG
A_Cariama_cristata PIFYIAVFALGLTGNLTVFFAIVKASSKKSITDIYLLSLAVSDILFVISLPFWASNVVRG
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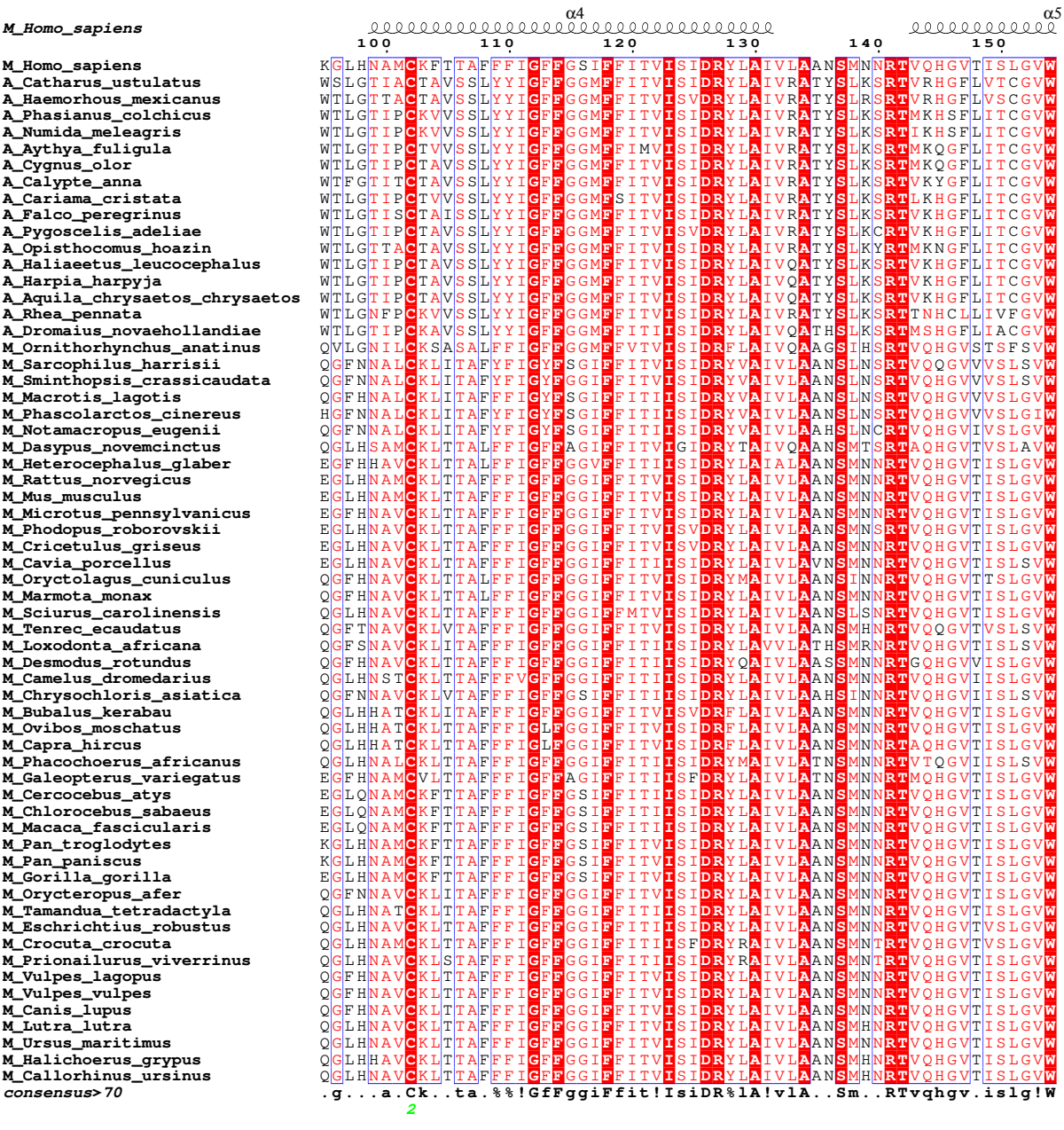
A_Pygoscelis_adeliae PVFYIAVFALGLTGNLMVGFAIVKEGSKKSITDIYLLNLAISDLLFVISLPFWASNTVRG
A_Opisthocomus_hoazin PIFYIAVFALGLAGNLMVVFAIVKEGNKKSITDFYLLNLAISDLLFVISLPFWASNTVRG
A_Haliaeetus_leucocephalus PIFYIAVFALGLTGNLMVVFAIMKEGSKKSITDIYLLNLAISDLLFVISLPFWASNTVRG
A_Harpia_harpyja PIFYIAVFALGLTGNLMVVFAIVKEGSKKSITDIYLLNLAISDLLFVISLPFWASNTVRG
A_Aquila_chrysaetos_chrysaetos PIFYIAVFALGLTGNLMVVFAIVKEGSKKSITDIYLLNLAISDLLFVISLPFWASNTVRG
A_Rhea_pennata PIFYTVVFALGLSGNLLVVFAIVKEGSIKSITDIYLLNLAVSDLLFVISLPFWAYSVVHG
A_Dromaius_novaehollandiae PIFYIAVFALGLTGNLLVVFAVVKEGSKKSITDIYLLNLAVSDLLFVISLPFWASNVVHG
M_Ornithorhynchus_anatinus PVLYALVFAFGLVGNLLVVFAVTNGGKKKSITDVYLLNLAFSDLLLVISLPFWIHYVVGE
M_Sarcophilus_harrisii PTLYSLVFAFGLVGNLLVVLALTTSSKHKSITDIYLLNLALSDLLFVATLPFWIHYVLNE
M_Sminthopsis_crassicaudata PTLYSLVFAFGLVGNLLVVLALTSSSKHKSITDIYLLNLALSDLLFVATLPFWTHYVLNE
M_Macrotis_lagotis PTLYSLVFAFGLVGNLLVVLALTNSSKHKSITDVYLLNLALSDLVFVTTLPFWTHYFINE
M_Phascolarctos_cinereus PTLYSLVFAFGLVGNMLVVLALTNSGKHKSITDIYLLNLALSDLLFVTTLPFWTHYVLNE
M_Notamacropus_eugenii PTLYSLVFAFGLVGNLLVVLALTNSSKHKSITDIYLLNLALSDLLFVTTLPFWTHYVLNE
M_Dasypus_novemcinctus ILLYSLVFVLGLAGNLLVLSALASSRKPRSVTDVYLLNLALSDLLFVATLPFWVQYMVSN
M_Heterocephalus_glaber SIFYSLIFTFGLVGNVLVVFALTRSQKSKSITDIYLLNLALSDLLFVATLPFWTHYLVSE
M_Rattus_norvegicus SIFYSLVFTFGLVGNLLVVLALTNSRKSKSITDIYLLNLALSDLLFVATLPFWTHYLISH
M_Mus_musculus SVFYALVFTFGLVGNLLVVLALTNSRKPKSITDIYLLNLALSDLLFVATLPFWTHYLISH
M_Microtus_pennsylvanicus SIFYSLVFAFGLVGNLLVVLALTNSRKPKSITDIYLLNLALSDLLFVATLPFWTHYLVSE
M_Phodopus_roborovskii SIFYSLVFAFGLVGNLLVVLALTNSRKPKSVTDIYLLNLALSDLLFVATLPFWTHYIVSE
M_Cricetulus_griseus SIFYSLVFAFGLVGNLLVVLALTNSRKPKSITDIYLLNLALSDLLFVATLPFWTHYVINE
M_Cavia_porcellus SIFYSLVFAFGLVGNLLVVFALTKSRKPKSITDIYLLNLALCDLLFVATLPFWTHYVINE
M_Oryctolagus_cuniculus SIFYSLVFAFGLVGNLLVVCALTSSRKPKSITDIYLLNLALSDLLFVATLPFWTHYVISE
M_Marmota_monax SIFYSVVFAFGLVGNLLVVMALNNSRKRKSITDIYLLNLALSDLLFVATLPFWTHYVVSE
M_Sciurus_carolinensis SIFYSLVFAFGLVGNLLVVFALTNSRKPKSITDIYLLNLALSDLLFVATLPFWTHYVLSE
M_Tenrec_ecaudatus PVFYSLVFAFGLVGNVLVVLALSNSRKAKSITDIYLLNLALSDLLFVATLPFWTHYVVSD
M_Loxodonta_africana PIFYSCVFAFGLVGNLLVVFVLTNSQKPKSITDIYLLNLAFSDLLFVATLPFWIHYVISD
M_Desmodus_rotundus PIVYSLVFAFGLVGNLLVVFALTHSQKPKSITDMYLLNLALSDLLFVVTLPFWIHYQVRE
M_Camelus_dromedarius AVLYSVVFACGLVGNLLVVFALTNSRKPKSITDIYLLNLALSDLLFVATLPFWTHYLINE
M_Chrysochloris_asiatica PIFYSFIFALGLVGNLLVVFSLTTSRKPKSITDIYLLNLALSDLLFVATLPFWIHYVIYD
M_Bubalus_kerabau VILYSLVFAFGLVGNLLVVFALINSQRSKSITDIYLLNLALSDLLFVATLPFWTHYVISE
M_Ovibos_moschatus ATLYSLVFAFGLVGNLLVVFALINSQRSKSITDIYLLNLALSDLLFVATLPFWTHYVINE
M_Capra_hircus ATLYSLVFAFGLVGNLLVVFALINSQRSKSITDIYLLNLALSDLLFVATLPFWTHYVINE
M_Phacochoerus_africanus SILYCLIFAFGLVGNLLVVFALTNSQKPKNITDIYLLNLALSDLLFVVTLPFWTHYLINE
M_Galeopterus_variegatus SIFYPLVFAFGLVGNLLVVFALVNTRKTKGIADIYLLNLALSDLLFVATLPFWTHYVLNE
M_Cercocebus_atys SIFYSVVFAIGLVGNLLVVFALTNSKKPKSVTDIYLLNLALSDLLFVATLPFWTHYVINE
M_Chlorocebus_sabaeus SIFYSVVFAIGLVGNLLVVFALTNSKKPKSVTDIYLLNLALSDLLFVATLPFWTHYVINE
M_Macaca_fascicularis SIFYSVVFAIGLVGNLLVVFALTNSKKPKSVTDIYLLNLALSDLLFVATLPFWTHYVINE
M_Pan_troglodytes SIFYSVVFAIGLVGNLLVVFALTNSKKPKSVTDIYLLNLALSDLLFVATLPFWTHYLINE
M_Pan_paniscus SIFYSVVFAIGLVGNLLVVFALTNSKKPKSVTDIYLLNLALSDLLFVATLPFWTHYLINE
M_Gorilla_gorilla SIFYSVVFAIGLVGNLLVVFALTNSKKPKSVTDIYLLNLALSDLLFVATLPFWTHYLINE
M_Orycteropus_afer PIFYSFVFAFGLVGNLLVVFALTHSRKPRSITDIYLLNLALSDLLFVATLPFWIHYVISD
M_Tamandua_tetradactyla PIFYSFVFAFGLVGNLLVVFALTNSRKPKSITDIYLLNLALSDLLFVATLPFWIHYVISE
M_Eschrichtius_robustus AILYSLVFAFGLVGNLLVVFALTNSQKPKSITDVYLLNLAFSDLLFVATLPFWTHYLINE
M_Crocuta_crocuta SILYSLVFAFGLVGNLLVVFALTNSRKPKSITDIYLLNLALSDLLFVATLPFWTHYLISE
M_Prionailurus_viverrinus SILYSLVFAFGLVGNLLVVFALTNSRKPKSITDIYLLNLALSDLLFVATLPFWTHYLMSE
M_Vulpes_lagopus SILYSLVFAFGLLGNLLVVFALTNSQKPKSITDIYLLNLALSDLLFVATLPFWTHYLMSE
M_Vulpes_vulpes SILYSLVFAFGLLGNLLVVFALTNSQKPKSITDIYLLNLALSDLLFVATLPFWTHYLMSE
M_Canis_lupus SILYSLVFAFGLLGNLLVVFALTNSQKPKSITDIYLLNLALSDLLFVATLPFWTHYLMSE
M_Lutra_lutra SILYSLVFAFGLLGNLLVVFALTNSQKPKSITDIYLLNLALSDLLFVATLPFWTHYLISE
M_Ursus_maritimus SILYSLVFAFGLLGNLLVVFALTNSRKPKSITDIYLLNLALSDLLFVATLPFWTHYLISE
M_Halichoerus_grypus SILYCLVFAFGLLGNLLVVFALTNSRKPKSITDIYLLNLALSDLLFAATLPFWTHYLISE
M_Callorhinus_ursinus SILYSLVFAFGLLGNLLVVFALTNSQKPKSITDIYLLNLALSDLLFVATLPFWTHYLLSE
consensus>70 .i.Y..vFa.GL.GNllVvfal..s...ks!tDiYLLnLA.sDllfv.tLPFW.hy.i.e


α4
α5
M_Homo_sapiens
100
110
120
130
140
150
M_Homo_sapiens
A_Catharus_ustulatus
A_Haemorhous_mexicanus
A_Phasianus_colchicus
A_Numida_meleagris
A_Aythya_fuligula
A_Cygnus_olor
A_Calypte_anna
A_Cariama_cristata
A_Falco_peregrinus
A_Pygoscelis_adeliae
A_Opisthocomus_hoazin
A_Haliaeetus_leucocephalus
A_Harpia_harpyja
A_Aquila_chrysaetos_chrysaetos
A_Rhea_pennata
A_Dromaius_novaehollandiae
M_Ornithorhynchus_anatinus
M_Sarcophilus_harrisii
M_Sminthopsis_crassicaudata
M_Macrotis_lagotis
M_Phascolarctos_cinereus
M_Notamacropus_eugenii
M_Dasypus_novemcinctus
M_Heterocephalus_glaber
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M_Mus_musculus
M_Microtus_pennsylvanicus
M_Phodopus_roborovskii
M_Cricetulus_griseus
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M_Oryctolagus_cuniculus
M_Marmota_monax
M_Sciurus_carolinensis
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M_Desmodus_rotundus
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M_Chrysochloris_asiatica
M_Bubalus_kerabau
M_Ovibos_moschatus
M_Capra_hircus
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M_Galeopterus_variegatus
M_Cercocebus_atys
M_Chlorocebus_sabaeus
M_Macaca_fascicularis
M_Pan_troglodytes
M_Pan_paniscus
M_Gorilla_gorilla
M_Orycteropus_afer
M_Tamandua_tetradactyla
M_Eschrichtius_robustus
M_Crocuta_crocuta
M_Prionailurus_viverrinus
M_Vulpes_lagopus
M_Vulpes_vulpes
M_Canis_lupus
M_Lutra_lutra
M_Ursus_maritimus
M_Halichoerus_grypus
M_Callorhinus_ursinus
consensus>70
2

β1
β2
η1
α6
α7
M_Homo_sapiens
160
170
180
190
200
210
M_Homo_sapiens
A_Catharus_ustulatus
A_Haemorhous_mexicanus
A_Phasianus_colchicus
A_Numida_meleagris
A_Aythya_fuligula
A_Cygnus_olor
A_Calypte_anna
A_Cariama_cristata
A_Falco_peregrinus
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M_Sarcophilus_harrisii
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M_Oryctolagus_cuniculus
M_Marmota_monax
M_Sciurus_carolinensis
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M_Loxodonta_africana
M_Desmodus_rotundus
M_Camelus_dromedarius
M_Chrysochloris_asiatica
M_Bubalus_kerabau
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M_Pan_paniscus
M_Gorilla_gorilla
M_Orycteropus_afer
M_Tamandua_tetradactyla
M_Eschrichtius_robustus
M_Crocuta_crocuta
M_Prionailurus_viverrinus
M_Vulpes_lagopus
M_Vulpes_vulpes
M_Canis_lupus
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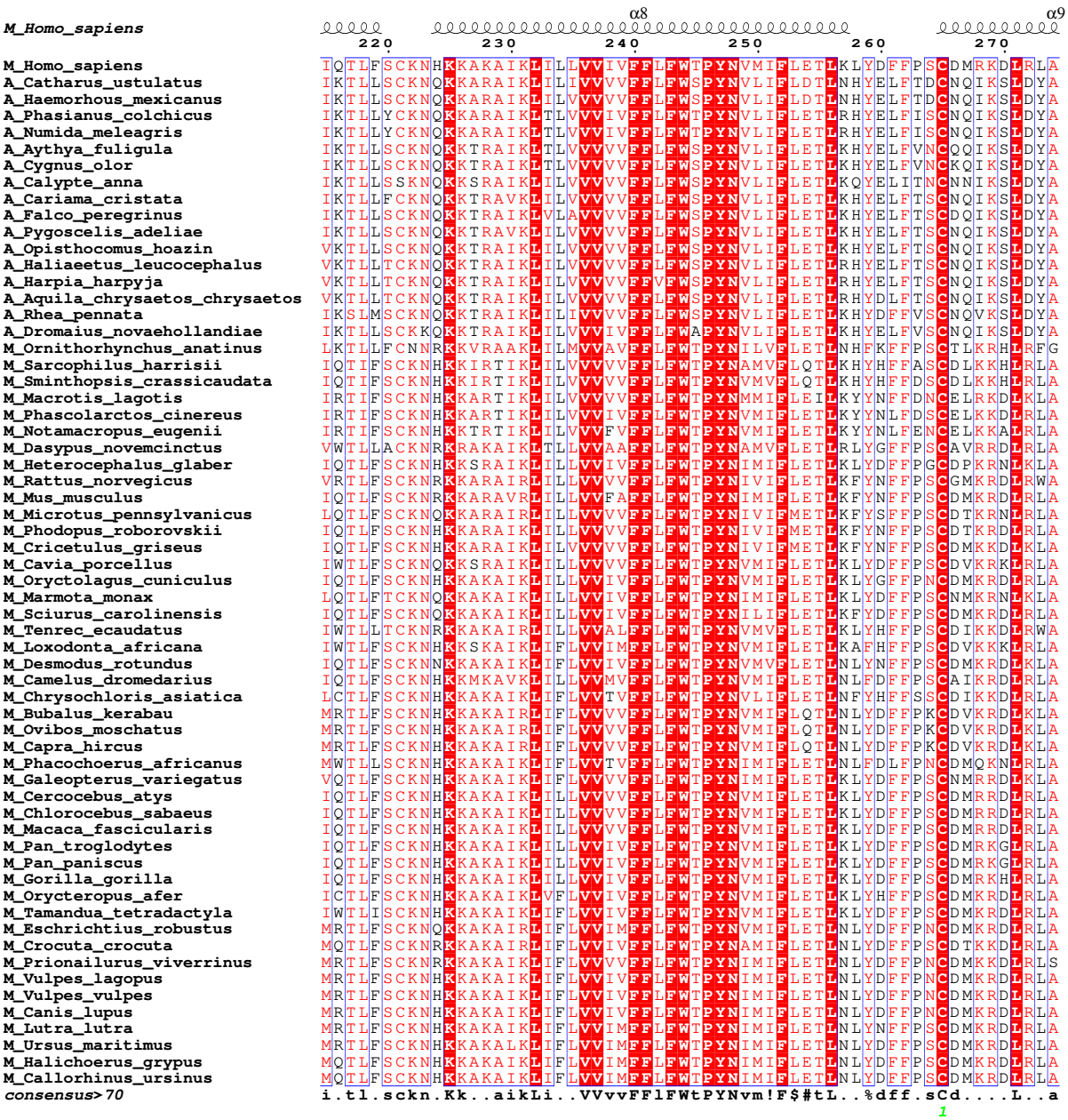
M_Ursus_maritimus
M_Halichoerus_grypus
M_Callorhinus_ursinus
consensus>70
2


α8
α9
M_Homo_sapiens
220
230
240
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260
270
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A_Phasianus_colchicus
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A_Opisthocomus_hoazin
A_Haliaeetus_leucocephalus
A_Harpia_harpyja
A_Aquila_chrysaetos_chrysaetos
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A_Dromaius_novaehollandiae
M_Ornithorhynchus_anatinus
M_Sarcophilus_harrisii
M_Sminthopsis_crassicaudata
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M_Microtus_pennsylvanicus
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M_Oryctolagus_cuniculus
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M_Vulpes_lagopus
M_Vulpes_vulpes
M_Canis_lupus
M_Lutra_lutra
M_Ursus_maritimus
M_Halichoerus_grypus
M_Callorhinus_ursinus
consensus>70
1

α10
α11
M_Homo_sapiens
280
290
300
310
320
330
M_Homo_sapiens LSVTETVAFSHCCLNPLIYAFAGEKFRRYLYHLYGKCLAVLCGRSVHVDFS..SSESQRS
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A_Haemorhous_mexicanus MHLAETIAFSHCCLNPFIYAFAGEKFRKYLYQVCLKYCPFLCFCGPCSHYQV...SHSVS
A_Phasianus_colchicus MHLTETIAFSHCCLNPLIYAFAGEKFRKYLYRVCFKYCPCLCFCGPCNRYEV...RPSTS
A_Numida_meleagris MHLTETVAFSHCCLNPFIYAFAGEKFRKYLHRVCFKYCPCLCFCGPCNTYQV...RPSAS
A_Aythya_fuligula MHLTETVAFSHCCLNPLIYAFAGEKFRKYLYSVCLKYCPCLCFCGPCSRYQV...SSSTS
A_Cygnus_olor MHLTETVAFSHCCLNPLIYAFAGEKFRKYLYSVCLKYCPCLCFCGPCSRYQV...SSSAS
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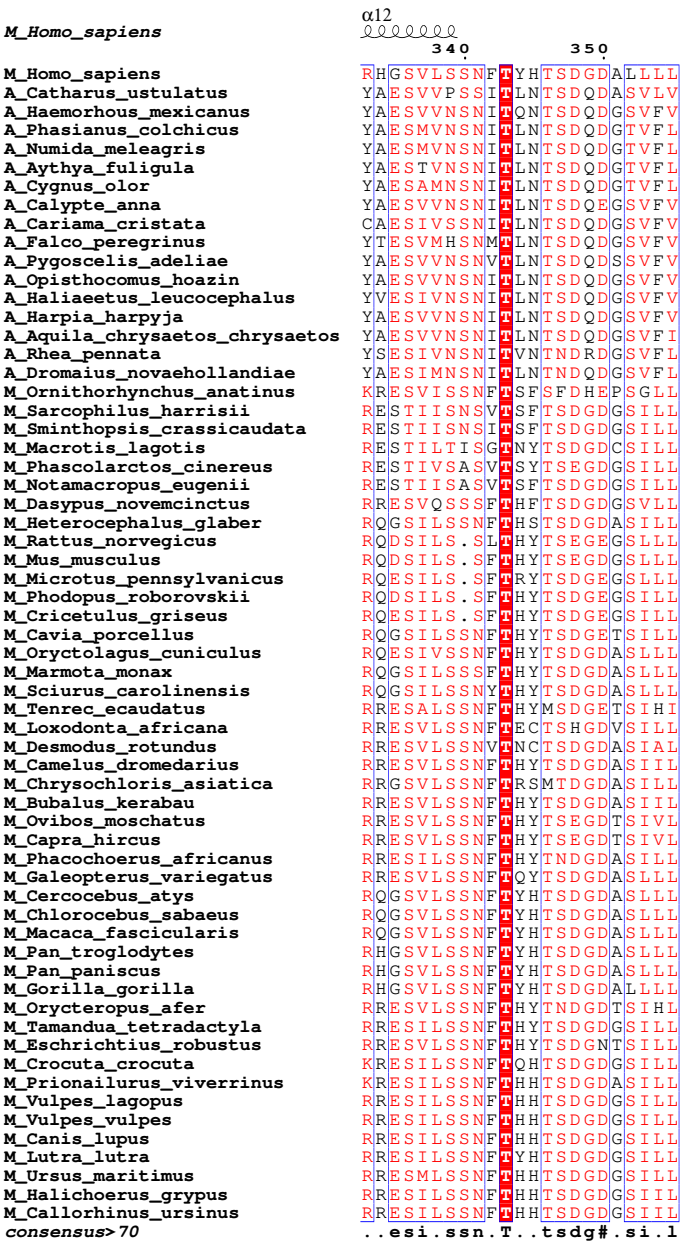
A_Cariama_cristata MHLTETIAFSHCCLNPLIYAFAGEKFRKYLYHVCLKHCPFLCFCGPCSRYQV...AYSAS
A_Falco_peregrinus MHLTETIAFSHCCLNPLIYAFAGEKFRKYLHHICLKYCPFLCFCGPCSHYQV...TSSAS
A_Pygoscelis_adeliae MHLTETIAFSHCCLNPFIYAFAGEKFRKYLYRVCLKYCPFLCFCGPCSRYQV...TYSAS
A_Opisthocomus_hoazin MHLTETIAFSHCCLNPLIYAFAGEKFRKYLYRVCLKYCPFLCFCGPCSRYQV...TYSAS
A_Haliaeetus_leucocephalus MHLTETIAFSHCCLNPFIYAFAGEKFRKYLYHLCLKYCPFLCFCGPCSRYQV...TYSAS
A_Harpia_harpyja MHLTETIAFSHCCLNPLIYAFAGEKFRKYLYHLCLKYCPFLCFCGPCSRYQV...TYSAS
A_Aquila_chrysaetos_chrysaetos MHLTETIAFSHCCLNPLIYAFAGEKFRKYLYHLCLKYCPFLCFCGPCSRYQV...TYSAS
A_Rhea_pennata MHLTETVAFSHCCLNPIIYAFAGKKFRKYLYQVCLKYCPFMCFCGPCNCYQV...SSSTS
A_Dromaius_novaehollandiae MHLTETVAFTHCCLNPVIYAFAGEKFRKYFYHVCLKYCPFMCFCGPCNRYQV...RSSTS
M_Ornithorhynchus_anatinus LSVTETIALIHCCLNPFIYAFAGKKFRRYLYVLYVKCRAVFCGLDPRLRIP..KEEVSLS
M_Sarcophilus_harrisii MNVTEAVALIHCCLNPFIYAFAGEKFRKYLYHLYRKCLSVLCNYDALSHIPA...PDAHK
M_Sminthopsis_crassicaudata MNVTEAVALIHCCLNPFIYAFAGEKFRKYLYHLYRKCLSVLCHYDALSQIPA...PDAHK
M_Macrotis_lagotis MTVTESIAFIHCCLNPFIYAFAGEKFRKYLYHLYRKCLSILCKHDVRTQFLP...TDTHK
M_Phascolarctos_cinereus MNVTETIAFIHCCLNPFIYAFAGEKFRKYLYHLYRKCLSILCSHDARSQFLS...PDAHK
M_Notamacropus_eugenii MNVTETIAFIHCCLNPFIYAFAGEKFRKYLYHLYRKCLSILCSHDVRTQFLS...PDAHK
M_Dasypus_novemcinctus LVVTETVAFSHCCLNPLIYAFAGEKFRKYLFRLCGKCLAVLCGRRVRGRFA..PSESQRS
M_Heterocephalus_glaber LNVTETLAFSHCCLNPLIYAFAGEKFRRYLYHLYRKCMVVLCGRSVHVTFS..RSESQKS
M_Rattus_norvegicus LSVTETVAFSHCCLNPFIYAFAGEKFRRYLRHLYNKCLAVLCGRPVHAGFS..T.ESQRS
M_Mus_musculus LSVTETVAFSHCCLNPFIYAFAGEKFRRYLGHLYRKCLAVLCGHPVHTGFS..P.ESQRS
M_Microtus_pennsylvanicus LSVTETVAFSHCCLNPLIYAFAGEKFRRYLGHLYRKCLAVLCCRPVHVNFT..P.ESQRS
M_Phodopus_roborovskii LSVTETVAFSHCCLNPFIYAFAGEKFRRYLGHLYRKCLAVMCGRSVHVDFT..P.ESQRS
M_Cricetulus_griseus LIVTETVAFSHCCLNPLIYAFAGEKFRRYLGHLYRKCLAVLCGRSVHVNFT..P.ESQRS
M_Cavia_porcellus LSVTETLAFSHCCLNPLIYAFAGEKFRRYLSHLYRKCKAALCGPSVHAGFS..ASESPRS
M_Oryctolagus_cuniculus LSVTETVAFSHCCLNPLIYAFAGQKFRRYLRHLSRKCQAVLCGRPVHVSFS..PSESQRS
M_Marmota_monax LNVTETVAFSHCCLNPLIYAFAGEKFRRYLCYLYRKYLAVLCGRSVHDSFS..PSESQRS
M_Sciurus_carolinensis LSVTETVAFSHCCLNPFIYAFAGEKFRRYLCHLYRKCLAVLCGRSVHVSFS..PPESQRS
M_Tenrec_ecaudatus LNVTETIAFSHCCLNPLIYAFAGEKFRKYLSHLYRKCLAVFCGRSLGMDLT..LSELQIS
M_Loxodonta_africana LSVTETIAFSHCCLNPLIYVFAGQKFRRHLYHLYRKYLAILCRHPVEVSFY..PSESQRS
M_Desmodus_rotundus LNVTETVAFIHCCLNPFIYAFAGEKFRRYLHHVYRKCLAVVCGGSVHASFP..ASESQRS
M_Camelus_dromedarius LSMTETIAFTHCCLNPLIYAFAGEKFRRYPYYLYRKCLAVLCCHPGHVTFSPSLSESQRS
M_Chrysochloris_asiatica LSVTETIAFSHCCLNPLIYAFAGEKFRKYLYYLYRKYLAVLCGCSVHVNLS..PPESQAS
M_Bubalus_kerabau INVTETIAFSHCCLNPLIYAFAGEKFRRYLYHLYRKCLAVLCCHPVHLSFSSSLSESQRS
M_Ovibos_moschatus ISVTETIAFSHCCLNPLIYAFAGEKFRRYLYHLYRKCLAVLCCRPVHITFSSSLSESQRS
M_Capra_hircus ISVTETIAFSHCCLNPLIYAFVGEKFRRYLYHLYRKCLAVLCCRPVHITFSSSLSESQRS
M_Phacochoerus_africanus LSVTETIAFTHCCLNPLIYAFAGEKFRRYLYHLYRKCLAILCGHPVTPSSS..LSESQRS
M_Galeopterus_variegatus LQVTETVAFSHCCLNPLIYAFAGEKFRRYLYHLYRKCLAVLCGHSVHISFP..PSESQRS
M_Cercocebus_atys LSVTETVAFSHCCLNPLIYAFAGEKFRRYLYHLYGKCLAVLCGRSVHVDFS..PSESQRS
M_Chlorocebus_sabaeus LSVTETVAFSHCCLNPLIYAFAGEKFRRYLYHLYGKCLAVLCGRSVHVDFS..PSESQRS
M_Macaca_fascicularis LSVTETVAFSHCCLNPLIYAFAGEKFRRYLYHLYGKCLAVLCGRSVHVDFS..PSESQRS
M_Pan_troglodytes LSVTETVAFSHCCLNPLIYAFAGEKFRRYLYHLYGKCLAVLCGRSVHVDFS..SSESQRS
M_Pan_paniscus LSVTETVAFSHCCLNPLIYAFAGEKFRRYLYHLYGKCLAVLCGRSVHVDFS..SSESQRS
M_Gorilla_gorilla LSVTETVAFSHCCLNPLIYAFAGEKFRRYLYHLYGKCLAVMCGRSVHVDFS..SSESQRS
M_Orycteropus_afer LSVTETIAFSHCCLNPLIYAFAGEKFRRYLYHLYRKCLAVLCGRSVHVSLS..PSESQTS
M_Tamandua_tetradactyla LSVTETVAFSHCCLNPLIYAFAGEKFRNYLYHLYGKCLAVLCGRPVHVSFS..PSETQRS
M_Eschrichtius_robustus LSVTETVAFTHCCLNPLIYAFAGEKFRRYLYHLYRKCLAVLCGHPVHVSFSLSPSETQRS
M_Crocuta_crocuta LSVTETIAFIHCCLNPFIYAFAGEKFRRYLYHLYRKCLAVLCGHPVHVSFS..PSE.QTS
M_Prionailurus_viverrinus LSVTETIAFTHCCLNPFVYAFAGEKFRRYLYHLCRKCRAVLCGRPVHVGFS..PSESQTS
M_Vulpes_lagopus LSVTETIAFTHCCLNPFIYAFAGEKFRRYLYHLYRKCLAVLCGHPVHVSFS..RTESQTS
M_Vulpes_vulpes LSVTETIAFTHCCLNPFIYAFAGEKFRRYLYHLYRKCLAVLCGHPVHVSFS..RTESQTS
M_Canis_lupus LSVTETIAFTHCCLNPFIYAFAGEKFRRYLYHLYRKCLAVLCGHPVHVSFS..RTESQTS
M_Lutra_lutra LSVTETIAFIHCCLNPFIYAFAGEKFRRYLNHLYRKCLAVLCGHPVHISFS..PTESQTS
M_Ursus_maritimus LSVTETIAFIHCCLNPFIYAFAGEKFRRYLYHLYRKCLAVLCGRPVHVSFS..PTESQTS
M_Halichoerus_grypus LSVTETIAFIHCCLNPFIYAFAGEKFRRYLYHLYRKCLAVLCGRPVHVSFS..PTESQTS
M_Callorhinus_ursinus LSVTETIAFIHCCLNPFIYAFAGEKFRRYLYHLYRKCLAVLCGRPVHVSFS..PTESQTS
consensus>70 l.vtEtiAf.HCCLNP.!YaFaGeKFR.ylyhl..K...vlC.......f.........s


α12
M_Homo_sapiens
340
350
M_Homo_sapiens RHGSVLSSNFTTYHTSDGDALLLL
A_Catharus_ustulatus YAESVVPSSITTLNTSDQDASVLV
A_Haemorhous_mexicanus YAESVVNSNITTQNTSDQDGSVFV
A_Phasianus_colchicus YAESMVNSNITTLNTSDQDGTVFL
A_Numida_meleagris YAESMVNSNITTLNTSDQDGTVFL
A_Aythya_fuligula YAESTVNSNITTLNTSDQDGTVFL
A_Cygnus_olor YAESAMNSNITTLNTSDQDGTVFL
A_Calypte_anna YAESVVNSNITTLNTSDQEGSVFV
A_Cariama_cristata CAESIVSSNITTLNTSDQDGSVFV
A_Falco_peregrinus YTESVMHSNMTTLNTSDQDGSVFV
A_Pygoscelis_adeliae YAESVVNSNVTTLNTSDQDSSVFV
A_Opisthocomus_hoazin YAESVVNSNITTLNTSDQDGSVFV
A_Haliaeetus_leucocephalus YVESIVNSNITTLNTSDQDGSVFV
A_Harpia_harpyja YAESVVNSNITTLNTSDQDGSVFV
A_Aquila_chrysaetos_chrysaetos YAESVVNSNITTLNTSDQDGSVFI
A_Rhea_pennata YSESIVNSNITTVNTNDRDGSVFL
A_Dromaius_novaehollandiae YAESIMNSNITTLNTNDQDGSVFL
M_Ornithorhynchus_anatinus KRESVISSNFTTSFSFDHEPSGLL
M_Sarcophilus_harrisii RESTIISNSVTTSFTSDGDGSILL
M_Sminthopsis_crassicaudata RESTIISNSITTSFTSDGDGSILL
M_Macrotis_lagotis RESTILTISGTTNYTSDGDCSILL
M_Phascolarctos_cinereus RESTIVSASVTTSYTSEGDGSILL
M_Notamacropus_eugenii RESTIISASVTTSFTSDGDGSILL
M_Dasypus_novemcinctus RRESVQSSSFTTHFTSDGDGSVLL
M_Heterocephalus_glaber RQGSILSSNFTTHSTSDGDASILL
M_Rattus_norvegicus RQDSILS.SLTTHYTSEGEGSLLL
M_Mus_musculus RQDSILS.SFTTHYTSEGDGSLLL
M_Microtus_pennsylvanicus RQESILS.SFTTRYTSDGEGSLLL
M_Phodopus_roborovskii RQDSILS.SFTTHYTSDGEGSILL
M_Cricetulus_griseus RQESILS.SFTTHYTSDGEGSILL
M_Cavia_porcellus RQGSILSSNFTTHYTSDGETSILL
M_Oryctolagus_cuniculus RQESIVSSNFTTHYTSDGDASLLL
M_Marmota_monax RQGSILSSSFTTHYTSDGDASLLL
M_Sciurus_carolinensis RQGSILSSNYTTHYTSDGDASLLL
M_Tenrec_ecaudatus RRESALSSNFTTHYMSDGETSIHI
M_Loxodonta_africana RRESVLSSNFTTECTSHGDVSILL
M_Desmodus_rotundus RRESVLSSNVTTNCTSDGDASIAL
M_Camelus_dromedarius RRESVLSSNFTTHYTSDGDASIIL
M_Chrysochloris_asiatica RRGSVLSSNFTTRSMTDGDASILL
M_Bubalus_kerabau RRESVLSSNFTTHYTSDGDASIIL
M_Ovibos_moschatus RRESVLSSNFTTHYTSEGDTSIVL
M_Capra_hircus RRESVLSSNFTTHYTSEGDTSIVL
M_Phacochoerus_africanus RRESILSSNFTTHYTNDGDASILL
M_Galeopterus_variegatus RRESVLSSNFTTQYTSDGDASLLL
M_Cercocebus_atys RQGSVLSSNFTTYHTSDGDASLLL
M_Chlorocebus_sabaeus RQGSVLSSNFTTYHTSDGDASLLL
M_Macaca_fascicularis RQGSVLSSNFTTYHTSDGDASLLL
M_Pan_troglodytes RHGSVLSSNFTTYHTSDGDASLLL
M_Pan_paniscus RHGSVLSSNFTTYHTSDGDASLLL
M_Gorilla_gorilla RHGSVLSSNFTTYHTSDGDALLLL
M_Orycteropus_afer RRESVLSSNFTTHYTNDGDTSIHL
M_Tamandua_tetradactyla RRESILSSNFTTHYTSDGDGSILL
M_Eschrichtius_robustus RRESVLSSNFTTHYTSDGNTSILL
M_Crocuta_crocuta KRESILSSNFTTQHTSDGDGSILL
M_Prionailurus_viverrinus KRESILSSNFTTHHTSDGDASILL
M_Vulpes_lagopus RRESILSSNFTTHHTSDGDGSILL
M_Vulpes_vulpes RRESILSSNFTTHHTSDGDGSILL
M_Canis_lupus RRESILSSNFTTHHTSDGDGSILL
M_Lutra_lutra RRESILSSNFTTYHTSDGDGSILL
M_Ursus_maritimus RRESMLSSNFTTHHTSDGDGSILL
M_Halichoerus_grypus RRESILSSNFTTHHTSDGDGSIIL
M_Callorhinus_ursinus RRESILSSNFTTHHTSDGDGSIIL
consensus>70 ..esi.ssn.T..tsdg#.si.l